

BANARAS HINDU UNIVERSITY

B.A. (Hons.) — Semester II Examination, 2023-24

Paper No. MTB-201: Calculus II

Time: 3 Hours

Full Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory — 2 marks each.)

- (a) Determine the repeated limits of $f(x, y) = \frac{x-y}{x+y}$ at $(0, 0)$.
- (b) By using the definition of limit, show that $\lim_{(x,y) \rightarrow (1,2)} xy = 2$.
- (c) If $z = x^2y^3$, where $x = t^2$ and $y = t$, then find $\frac{dz}{dt}$.
- (d) Investigate for maximum and minimum values the function given by $f(x) = \sin x + \cos 2x$.
- (e) If $f(x, y) = \begin{cases} \frac{xy}{\sqrt{x^2+y^2}}, & \text{if } x^2 + y^2 \neq 0 \\ 0, & \text{if } x = y = 0 \end{cases}$, then find $f_x(0, 0)$ and $f_y(0, 0)$.
- (f) Verify Euler's theorem for the function $u = x^3 \log(y/x)$.
- (g) Show that $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$.
- (h) If $u^u v^v w^w = 1$, then show that $\frac{\partial^2 u}{\partial v \partial w} = -(u \log(eu))^{-1}$ when $u = v = w$.
- (i) Show that $\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$.
- (j) Evaluate: $\int_1^e \int_0^{\log y} \int_1^{e^x} \log z dz dx dy$.

Question 2.

- (a) Show that the function $f(x, y) = \sqrt{|xy|}$, where $(x, y) \in \mathbb{R}^2$, is continuous but not differentiable at the origin. [6]
- (b) State and prove a sufficient condition for differentiability of a function of two variables. [6]

Question 3.

- (a) State and prove Schwarz's theorem for a function of two variables. [6]
- (b) Expand the function $f(x, y) = x^2y + 3y - 2$ by Taylor's theorem in powers of $(x - 1)$ and $(y + 2)$. [6]

Question 4.

(a) If $u = \tan^{-1} \left(\frac{xy}{\sqrt{1+x^2+y^2}} \right)$, then show that $\frac{\partial^2 u}{\partial x \partial y} = \frac{1}{(1+x^2+y^2)^{3/2}}$. [6]

(b) State and prove Euler's theorem for homogeneous functions. If $u = \tan^{-1} \left(\frac{x^3+y^3}{x-y} \right)$, $x \neq y$, then show that:

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$$

[6]

Question 5.

(a) Determine the points where the function $f(x, y) = x^4 + y^4 - 2x^2 + 4xy - 2y^2$ has a maximum or minimum. [6]

(b) Prove that of all rectangular parallelepipeds of a given volume, the cube has the least surface area. [6]

Question 6.

(a) Change the order of integration in the double integral:

$$\int_0^{2a} \int_{\sqrt{2ax-x^2}}^{\sqrt{2ax}} f(x, y) dy dx \quad (a > 0)$$

[6]

(b) Transform the integral $\int_0^1 \int_0^{\sqrt{1-x^2}} y^2(x^2 + y^2) dy dx$ by changing to polar coordinates, and hence evaluate it. [6]

Question 7.

(a) State and prove Legendre's duplication formula. [6]

(b) Evaluate:

$$\iiint x^{l-1} y^{m-1} z^{n-1} dx dy dz$$

where x, y, z are always positive but limited by the condition $x + y + z \leq 1$. [6]

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